

Library Management System (LMS)

Phase 1: Project Proposal & Requirements

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Chapter One

Project Proposal and Requirements Phase

1.1 Chosen System

Library Management System (LMS)

1.2 Problem Definition

1.2.1 The Current System or Scenario

Currently, the library operates using a fragmented, manual system. Inventory tracking relies on physical card catalogs and decentralized spreadsheets, while member registrations and daily transactions (borrowing and returning books) are recorded in paper-based ledgers. Librarians must manually check shelves or disconnected files to verify book availability and stamp due dates by hand.

1.2.2 Problems that Require a Database Solution

The existing manual setup introduces several operational bottlenecks:

- **Data Redundancy and Inconsistency:** Member details and book records are duplicated across multiple sheets, leading to conflicting information.
- **Inefficient Search Operations:** Locating a specific book by Title, Author, or ISBN requires manual scanning, which is time-consuming and prone to human error.
- **Poor Transaction Tracking:** Identifying overdue books and calculating fines is a completely manual process, resulting in lost inventory and uncollected penalties.
- **Lack of Concurrency:** Spreadsheets do not efficiently support simultaneous multi-user access by different library staff.

1.2.3 Benefits of Implementing the Database

Transitioning to a robust Relational Database Management System (RDBMS) provides the following advantages:

- **Centralized Data:** A normalized database eliminates redundancy, ensuring member and inventory data is stored once and updated globally.
- **Automated Tracking:** Borrowing, returning, and fine calculations are handled programmatically, ensuring accuracy and accountability.
- **Rapid Search and Retrieval:** Indexed tables allow instantaneous querying of the catalog, drastically reducing the time needed to serve members.
- **Data Integrity and Security:** The database enforces constraints to prevent invalid data entry and restricts access to sensitive member information.

1.3 System Requirements

1.3.1 Functional Requirements

The functional requirements define what the system will do:

- **Add, Update, and Delete Records:** Librarians must be able to seamlessly manage book inventory (add new arrivals, update details, remove lost books) and member profiles.
- **Search Operations:** The system must allow users to search the library catalog by ISBN, Title, Author, Category, or Publication Year.
- **Manage Transactions:** The system must process book check-outs, returns, and renewals, automatically updating the available copies count.
- **Generate Reports:** The system must generate automated reports, including lists of currently active loans, overdue books, accumulated fines, and total inventory statistics.

1.3.2 Non-Functional Requirements

The non-functional requirements define the performance and quality standards the system must meet:

- **Data Security:** The system shall implement Role-Based Access Control (RBAC). Only authorized librarians may execute modifications (INSERT, UPDATE, DELETE), while sensitive data like passwords must be hashed.
- **Performance:** The database must be optimized using proper indexing so that standard search queries and transaction updates execute in under two seconds.
- **Reliability:** The system must guarantee ACID (Atomicity, Consistency, Isolation, Durability) properties to ensure that a book check-out transaction cannot be partially completed or corrupted.
- **Backup and Recovery:** The database must support automated daily backups, allowing administrators to restore the system to a previous state in the event of hardware failure or accidental data loss.